CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) — CHAPTER 6
LINKING SPREADSHEET DATA

Comprehensive Detailed Theory • Unabridged Practical Procedures • Cross-Sheet Referencing • Hyperlinks & External Data • Solved Board PYQs & Case Studies

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1. Need for Multiple Worksheets & Workbook Organization

In professional financial and operational environments, a single worksheet is insufficient to manage complex, interrelated data. Segregating information across multiple worksheets offers substantial structural advantages:

- **Organizational Clarity:** Enables keeping distinct operational entities separated (e.g., maintaining individual monthly expense sheets for regional branch offices like *Branch1*, *Branch2*, *Branch3*, or separate sheets for individual bank accounts).
- **Consolidation & Centralized Reporting:** A master summary worksheet can dynamically reference, aggregate, and calculate totals from all individual branch or account sheets.
- **Default Workbook Setup:** By default, when a new spreadsheet workbook is created in LibreOffice Calc, it contains sheets named **Sheet1** (or up to 3 sheets in standard editions: *Sheet1*, *Sheet2*, *Sheet3*). These are managed via **Sheet Tabs** positioned at the bottom left corner of the Calc window.
- **Active Sheet Identification:** The currently visible and active worksheet tab is highlighted/underlined on the tab bar. Clicking any sheet tab instantly makes it active.
- **Standard File Extension:** Workbooks created in LibreOffice Calc are saved with the standard OpenDocument Spreadsheet file extension: **.ods**.

2. Setting Up & Managing Multiple Sheets (Unabridged Procedures)

A. Unabridged Step-by-Step Procedure to Insert a New Sheet:

Step 1: Select the existing sheet that will be positioned immediately before or after the new sheet.

Step 2: Choose any of the following methods to trigger the insert dialog:
- Method 1 (Menu Bar): Choose **Sheet** → **Insert Sheet...**.
- Method 2 (Context Menu): Right-click on any **Sheet Tab** at the bottom and select **Insert Sheet...** from the pop-up menu.
- Method 3 (Plus Icon): Click the **+ (Add Sheet)** icon located just before the sheet tabs.

Step 3: The **Insert Sheet** dialog box appears on the screen.

Step 4: Under the **Position** section, choose either **Before current sheet** or **After current sheet**.

Step 5: Under the **Sheet** section:
- To insert a blank sheet, ensure **New sheet** is selected. Specify the **No. of sheets** to add (default is 1). In the **Name** box, type a customized name (e.g., *Sales_Datasheet*) or retain the default *Sheet<n>*.
- To insert from an existing external spreadsheet, select **From file**, click **Browse...**, choose the file, and click Open. (Check the *Link* box if you want changes in the external file to dynamically reflect in your workbook).

Step 6: Click **OK**. The new worksheet is inserted at the chosen position.

B. Unabridged Step-by-Step Procedure to Rename a Sheet:

Step 1: Select the sheet you wish to rename.

Step 2: Use any of the following three ways:
- **Double-click** directly on the sheet tab at the bottom.
- Right-click the sheet tab and select **Rename Sheet...** from the context menu.
- Navigate to the Menu bar and choose **Sheet** → **Rename Sheet...**.

Step 3: In the pop-up **Rename Sheet** dialog box, delete the existing name and type the new name (e.g., *Q1_Summary*).

Step 4: Click **OK**. The tab label updates immediately.

C. Unabridged Step-by-Step Procedure to Delete a Sheet:

Step 1: Right-click on the tab of the sheet you want to remove and select **Delete Sheet...** (or select the sheet and choose **Sheet** → **Delete Sheet...** from the Menu bar).

Step 2: A confirmation dialog appears asking: *'Are you sure you want to delete the selected sheet(s)?'*.

Step 3: Click **Yes** to permanently remove the sheet and its contents from the workbook.

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3. Referencing Data Across Sheets (Within the Same Workbook)

Definition: A **Cell Reference** refers to a specific cell or range of cells whose values are used in formulas. When referencing a cell located on a *different sheet* within the same workbook, Calc requires a **qualified cell reference**.

The Two Components of a Cross-Sheet Reference (CBSE Board Exam Must-Know):

- **1. Sheet Name:** The exact name of the source worksheet followed by a **dot (.) separator**.
- **2. Cell Address / Coordinates:** The column letter and row number identifying the cell on that sheet.
- **Standard Syntax:** SheetName.CellAddress *(e.g., Sheet2.A1 or $Sheet2.$A$1)*.
- **Space Handling in Sheet Names:** If the source sheet name contains spaces, it must be enclosed within single quotes: 'Sales Data'.B4 or 'My Sheet'.C10.

A. Referencing via Keyboard:

Type the qualified formula directly into the formula bar. For example, to add cell *A1* of *Sheet2* and cell *B4* of *Sheet3* inside cell *A1* of *Sheet1*, type:
=Sheet2.A1 + Sheet3.B4
When the values in *Sheet2.A1* or *Sheet3.B4* are edited, Calc automatically recalculates the sum in *Sheet1.A1*.

B. Referencing via Mouse (Point-and-Click Method):

Step 1: Click on the destination cell where the formula result should appear (e.g., cell *C4* in *Sheet5*).

Step 2: Type the equal sign (**=**) or click the Function (**=**) icon on the Formula bar.

Step 3: Type the first part of your mathematical expression (e.g., C2 - ( ).

Step 4: Click the sheet tab at the bottom for the first source sheet (e.g., *Sheet3*). The sheet opens. Click directly on cell *B15*. Calc automatically appends Sheet3.B15 to the formula line.

Step 5: Type the mathematical operator (e.g., **+**) in the formula bar.

Step 6: Click the sheet tab for the second source sheet (e.g., *Sheet4*) and click directly on cell *B9*. Calc appends Sheet4.B9.

Step 7: Type the closing parenthesis) and press **Enter**. The active focus returns to *Sheet5*, displaying the evaluated result.

4. Hyperlinks in Spreadsheets (Concepts, Types & Navigation)

Definition: A **Hyperlink** in a spreadsheet is an interactive reference or navigational link. Clicking a hyperlink (or pressing **Ctrl + Click**) navigates the user directly to a specific target—such as a website, another file, or a specific sheet/cell within the current workbook.

Why Use Hyperlinks in Spreadsheets?

- **Prevents Document Bloat:** Keeps workbook size manageable by linking to supporting files or documentation rather than embedding massive redundant content.
- **Rapid Navigation:** Facilitates instant jumps between disparate sheets, named ranges, or database tables across large corporate workbooks.
- **Access to Live External Resources:** Directly connects spreadsheet models to live web resources, intranets, or shared network folders.

The Four Hyperlink Categories (Hyperlink Dialog Box Left Pane):

Category	Icon / Option	Technical Purpose & Destination
Internet	Globe icon	Creates hyperlinks to World Wide Web URLs (HTTP/HTTPS) or FTP servers. Can target specific web anchors.
Mail & News	Envelope icon	Creates a mailto link to open the default email client with pre-filled recipient address and subject line.
Document	Document page icon	Creates hyperlinks to an existing external file OR targets a specific sheet/range in the *current document*.
New Document	Blank page with star	Creates a hyperlink that simultaneously creates a brand-new file (text document, spreadsheet, presentation) when clicked.

Unabridged Step-by-Step Procedure to Insert a Document Hyperlink:

Step 1: Click inside the cell where you want to insert the hyperlink.

Step 2: Click the **Hyperlink** icon on the Standard Toolbar or choose **Insert** → **Hyperlink...** (or press **Ctrl + K**).

Step 3: In the left pane of the Hyperlink dialog, click on **Document**.

Step 4: In the **Path** field, click the folder icon to browse and select the external target file from your computer (e.g., *annual_report.ods*). If linking to a location inside the *current document*, leave the Path field blank.

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Step 5: Target in Document Feature: Click the crosshair target icon next to the *Target* box. The *Target in Document* tree opens displaying all **Sheets**, **Range names**, and **Database ranges**. Expand *Sheets*, select the specific sheet (e.g., *House_Loan_25yr*), and click **Apply**.

Step 6: Under **Further Settings**:
- **Form:** Select **Text** (to display as clickable colored text) or **Button** (to display as an interactive form button control).
- **Text:** Enter the visible link label (e.g., *'Jump to 25 Year Loan Summary'*).

Step 7: Click **Apply**, and then click **OK**. Calc inserts the formatted hyperlink.

Editing or Removing a Hyperlink:

To edit an existing hyperlink, click the cell to position the cursor without following the link, click **Insert** → **Hyperlink...** (or click the Hyperlink icon on toolbar), alter the path, target, or text, and click Apply. To remove a hyperlink completely while keeping the text, right-click the hyperlinked cell and choose **Remove Hyperlink**.

5. Relative vs. Absolute Hyperlinks (High-Yield CBSE Concept)

Hyperlink destinations are recorded using file path strings. The distinction between relative and absolute addressing is crucial when files are shared, moved, or uploaded across computer networks:

A. Absolute Hyperlinks:

- **Definition:** An absolute hyperlink defines the complete, full path required to locate a file starting from the top-most root directory or drive letter.
- **Structure:** Begins from the root: e.g., file:///D:/CBSE/IT402/Spreadsheets/Report.ods or https://www.cbse.gov.in/index.html.
- **Portability Behavior: Breaks when files are moved to another computer!** If Arjun creates an absolute link on Drive D: of his home PC and emails the project to his teacher, the link fails because the teacher's PC lacks that exact directory hierarchy on Drive D:.
- **Stability:** An absolute link stops working *only if the target destination file itself is moved or deleted*.

B. Relative Hyperlinks:

- **Definition:** A relative hyperlink defines the path to the target document *with respect to the current location (working directory)* of the active document.
- **Structure & Notation:**
  - Single dot (**.**) represents the *current folder*: e.g., .\data.ods.
  - Double dots (**..**) represent the *parent folder* (one level up): e.g., .. eports\summary.ods.
- **Portability Advantage: Highly portable!** If an entire project folder containing both the source workbook and linked files is copied or emailed to another PC, relative hyperlinks remain fully operational as long as their relative folder hierarchy is preserved.
- **Break Condition:** A relative link stops working only if the source and target files are moved relative to each other.

Comparison: Relative vs. Absolute Hyperlinks:

Comparison Parameter	Relative Hyperlink	Absolute Hyperlink
Starting Point	Starts from the **current location** of the open document.	Starts from the **top-most root directory** / drive letter.
Path Representation	Uses relative folder notation (e.g., ..\ss1.ods or .\ss3.ods).	Full path starting from root (e.g., D:\Main\Second\ss3.ods).
Portability Across Systems	**Maintains links** when entire directory structure is moved to another computer.	**Fails / Breaks** when shared or copied to a computer with different drives/paths.
When Link Breaks	Breaks only if source and target files are moved relative to each other.	Breaks if the target file is renamed, moved, or drive letter changes.
Calc Display Default	Can be saved relatively, but Calc displays URLs in **absolute form** in the dialog.	Always displayed in full absolute form (e.g., file:///D:/...).

6. Linking to External Data (Web Pages & HTML Tables)

LibreOffice Calc allows inserting and maintaining live links to external tabular data located in HTML files, web pages, or other spreadsheet files. When the source web page or HTML file updates, Calc refreshes the linked table automatically.

Role of the External Data Dialog Box (CBSE 2-Marker):

The *External Data* dialog box allows users to specify the URL or file path of an external data source, preview and select specific named tables or ranges (e.g., HTML_1, HTML_all), and set an automatic refresh frequency (**Update every [n] seconds**).

Unabridged Step-by-Step Procedure to Link to External Web Data:

Step 1: Open the Calc worksheet where the external table data is to be inserted (the target document).

Step 2: Select the cell that will serve as the top-left anchor for the incoming table data.

Step 3: Choose **Sheet** → **External Links...** from the Menu bar. The *External Data* dialog box opens.

Step 4: In the **URL of External Data Source** field, enter the full web address (e.g., https://www.example.gov/stats.html) or click **Browse...** to locate a local HTML or spreadsheet file. Press **Enter** after typing or selecting the path.

Step 5: If prompted by the *Import Options* dialog (for web/text formatting), leave the locale setting as *Automatic* and click **OK**.

Step 6: Calc queries the source and populates the **Available Tables/Ranges** listbox (displaying elements like *HTML_all*, *HTML_1*, *HTML_2*). Highlight the specific table(s) you wish to link. (Hold **Ctrl** to select multiple tables).

Step 7: Automatic Refresh Option: Check the box **'Update every'** and specify the refresh interval in seconds (e.g., update every 60 seconds) so live data updates continuously.

Step 8: Click **OK**. Calc imports the table data and establishes a dynamic link.

7. Linking to Registered Data Sources in LibreOffice Calc

Definition: In LibreOffice, a **Registered Data Source** refers to an external database (such as a LibreOffice Base database file in **.odb** format, or an external database like MySQL/PostgreSQL registered via ODBC/JDBC) that has been configured in LibreOffice's global database catalog. Once registered, any Calc spreadsheet can easily access, view, and pull records from its tables and queries.

- **File Format:** Natively saved as **.odb** (OpenDocument Database).
- **Viewing Shortcut (F4):** Pressing **F4** (or choosing **View** → **Data Sources**) toggles open the **Data Source View pane** above the spreadsheet.
- **Data Explorer Structure:** The left panel displays all registered database connections (including the built-in *Bibliography* database). Expanding the tree shows **Tables** and **Queries**.
- **Two Ways to Insert Records into Calc:**
  1. **Drag-and-Drop:** Drag individual field headers or specific records directly onto spreadsheet cells.
  2. **Data to Text Tool:** Highlight multiple rows using the row selector buttons in the database table grid, and click the **Data to Text** icon on the toolbar above the pane. Calc pastes the selected records into the active worksheet with column headers.

8. CBSE Board Examination Question Bank & Model Marking Scheme Solutions

Part A: Objective & Multiple Choice Questions (1 Mark Each)

Q1. When you open a new spreadsheet, by default, it has a sheet named _____ which is managed using tabs at the bottom of the spreadsheet. [CBSE SQP 2022-23]
(a) Sheet1 (b) Untitled1 (c) Worksheet1 (d) New Sheet

Ans: (a) Sheet1

Q2. In a spreadsheet using to create a hyperlink to a web FTP or Telnet, click on the _____ icon available in Hyperlink dialog box. [CBSE SQP 2022-23]
(a) browser (b) hyperlink (c) Internet (d) mail & news

Ans: (c) Internet (represented by the globe icon in the left pane of the Hyperlink dialog).

Q3. A _____ refers to a cell or a range of cells on a worksheet and can be used to find the values or data that you want formula to calculate. [CBSE SQP 2022-23 / Board 2023]
(a) Cell reference (b) Block (c) Sheet Reference (d) Autofill

Ans: (a) Cell reference

Q4. Identify the part of a Spreadsheet using which a user can manage multiple sheets. [CBSE SQP 2023-24]
(a) Status Bar (b) Sheet Tab (c) Formula Bar (d) Worksheet Area

Ans: (b) Sheet Tab

Q5. At the bottom of each worksheet, left tab indicates the _____ of the worksheet in a workbook. [CBSE SQP 2023-24]
(a) Size (b) Type (c) Name (d) Color

Ans: (c) Name

Q6. A _____ hyperlink stores the location with respect to the current location in LibreOffice Calc. [CBSE SQP 2024-25 / 2025-26 / 2026-27]
(a) Absolute (b) Relative (c) Mixed (d) Reference

Ans: (b) Relative

Q7. In LibreOffice Calc, to refer to a cell in another sheet of the same file, precede the cell reference with a _____ sign. [CBSE Board Paper 2 (Set 4)]
(a) $ (b) # (c) . (dot) (d) !

Ans: (c) . (dot) (Syntax: SheetName.CellReference, e.g., Sheet2.A1).

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Q8. In Calc, which of the following types of hyperlink uses the complete location where the file to be linked is stored? [CBSE Board Paper 1 (Set 4)]
(a) Absolute hyperlink (b) Mixed hyperlink (c) Relative hyperlink (d) Static hyperlink

Ans: (a) Absolute hyperlink

Q9. The extension of the file created in LibreOffice Calc is: [CBSE Board Paper 2 (Set 4)]
(a) .odg (b) .odt (c) .odc (d) .ods

Ans: (d) .ods (OpenDocument Spreadsheet).

Part B: Short Answer Questions (2 Marks Each)

Q10. What do you mean by Hyperlinks in Spreadsheets? Give the two different types of Hyperlinks that can be used in Spreadsheets. [CBSE SQP 2022-23, Q.13]

Ans: A hyperlink is an interactive clickable reference that opens another document, website, or navigates to a specific sheet/cell in the active workbook.
The two primary types based on path addressing are:
1. **Absolute Hyperlink** (defines the complete path starting from root).
2. **Relative Hyperlink** (defines the path relative to the current file's directory).

Q11. What are the two parts of a cell reference while referencing data on other sheets? Explain with an example. [CBSE Board 2024, Q.14]

Ans: When referencing data on other sheets, a qualified cell reference consists of two distinct parts separated by a dot (**.**):
1. **Sheet Name:** The exact name of the source worksheet (e.g., Sheet2).
2. **Cell Address:** The column letter and row number identifying the cell (e.g., A1).
Example: In the expression =Sheet2.A1, *Sheet2* is the sheet name and *A1* is the cell address.

Q12. What is the role of the External Data Dialog box in LibreOffice Calc? Briefly discuss. [CBSE Board Paper 2 (Set 4), Q.13]

Ans: The *External Data* dialog box (opened via **Sheet** → **External Links...**) is used to import and establish live dynamic links to external data tables from HTML files, web pages, or other spreadsheets. It allows selecting specific named tables/ranges (e.g., HTML_1) and setting an automatic refresh interval (**Update every 'n' seconds**) to keep spreadsheet data synchronized with live sources.

Part C: Descriptive, Long Answer & Case-Study Questions (4 Marks Each)

Q13. Distinguish between the following with respect to spreadsheets: [CBSE Board 2023, Q.18]
(a) Relative and Absolute hyperlink.
(b) Internet hyperlink and Document hyperlink.

Ans:
(a) Relative vs. Absolute Hyperlink:
• *Relative Hyperlink:* Stores the path to the target document relative to the current folder location (e.g., ..\data.ods). It remains functional when the entire folder structure is moved to another computer.
• *Absolute Hyperlink:* Stores the complete URL or directory path from the root drive (e.g., D:\Work\data.ods). It breaks if the files are transferred to a system with a different drive/folder path.
(b) Internet vs. Document Hyperlink:
• *Internet Hyperlink:* Targets web pages or FTP servers across the Internet using URLs starting with http:// or https://.
• *Document Hyperlink:* Targets an existing local file on the computer or connects to a specific sheet, cell range, or bookmark within the *same* spreadsheet document.

Q14. Explain four types of hyperlinks that can be applied in spreadsheets. [CBSE Board 2024, Q.18]

Ans: In LibreOffice Calc, the Hyperlink dialog box provides four distinct hyperlink categories:
1. **Internet:** Creates a link to a website URL (HTTP/HTTPS) or FTP location.
2. **Mail & News:** Creates a clickable email link (mailto) with pre-set recipient and subject.
3. **Document:** Links to an existing external document on the computer or navigates to a specific internal target (sheet, named range) within the current workbook.
4. **New Document:** Creates a link that generates a new document file immediately or edits it later when clicked.

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Q15. Case Study: Arjun is preparing a project report in LibreOffice Calc that contains multiple spreadsheets. He wants smooth navigation between different sheets and also plans to include data from a government website. Initially, he uses an absolute hyperlink but later realizes it breaks when he shares the report with his teacher because the file path is different on another computer. His friend suggests trying a relative hyperlink instead. While inserting a hyperlink, Arjun notices the option 'Target in Document', which allows him to connect directly to a particular sheet. He also discovers the 'Link to External Data' option to fetch live tables from a web page into his spreadsheet. [CBSE SQP 2024-25 / 2025-26 / 2026-27, Q.21]
(a) Why did the absolute hyperlink fail when Arjun shared the file, and how does a relative hyperlink solve this problem?
(b) How does the 'Target in Document' option increase the usefulness of hyperlinks in large spreadsheets?
(c) Arjun wants to ensure that anyone viewing his report can update data directly from a government website. Which feature should he use, and why is it beneficial?
(d) If Arjun wants to edit or remove a hyperlink later, what steps should he take?

Ans:
(a) The absolute hyperlink failed because it contained hardcoded full drive paths (e.g., pointing to drive D:\... on Arjun's computer) which did not exist on the teacher's computer. A **relative hyperlink** solves this problem because it points to the file based on the relative position between the files. As long as the project folder containing the files is shared together, relative links continue working regardless of drive names.
(b) The **Target in Document** option allows hyperlinks to point directly to a specific sheet, named cell range, or database range within the workbook. In large multi-sheet workbooks, this eliminates tedious manual scrolling and provides instant one-click navigational jumps.
(c) He should use the **Link to External Data** feature (via **Sheet** → **External Links...**) with the **'Update every [n] seconds'** option enabled. It is beneficial because it dynamically fetches and refreshes live web tables into the spreadsheet automatically without manual copying.
(d) To edit or remove a hyperlink:
• *To Edit:* Select the cell without triggering the link, click the **Hyperlink** icon (or **Insert** → **Hyperlink**), alter the target path/text, and click **Apply**.
• *To Remove:* Right-click the hyperlinked cell and select **Remove Hyperlink** from the context menu (or select the cell and press Delete).

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